

2-Bus Loadflow

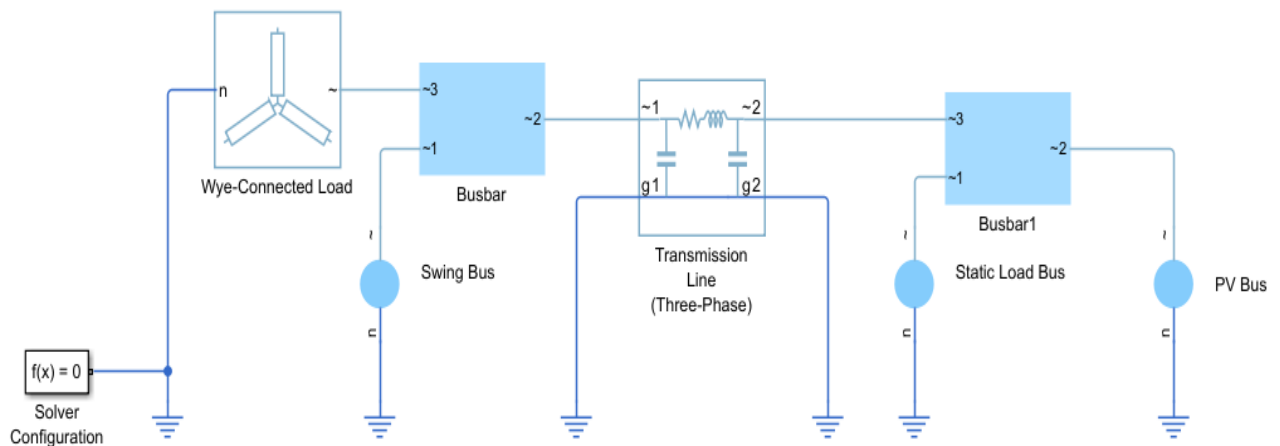
Simulink/Simspace

Introduction

This project presents a **2-Bus Loadflow model** developed in MATLAB/Simulink to analyze steady-state power system behaviour. The objective of the model is to study how active and reactive power flow between buses is influenced by network components such as loads, generators, and transmission lines. The simulation helps evaluate voltage profiles, power transfer capability, and system stability under typical operating conditions.

The system consists of two main buses: a **Swing Bus** and a **PV Bus**, interconnected through a three-phase transmission line. The model incorporates realistic elements including a wye-connected load, static load, busbars, and generator representations, providing essential insight into fundamental power-flow analysis.

2-Bus Loadflow



Component Descriptions

1. Solver Configuration

This block defines the numerical settings required for Simscape simulations. It ensures that electrical components modeled in the Simulink environment operate with consistent solver parameters, enabling stable and accurate power-flow results.

2. Wye-Connected Load

A three-phase load connected in a wye configuration. It represents a balanced consumer load drawing active and reactive power from the system. This load is connected to the Swing Bus through Busbar 0.

3. Swing Bus (Reference Bus)

The Swing Bus serves as the voltage and angle reference for the entire power system. It automatically balances real and reactive power mismatches in the network. This bus defines the system's base voltage magnitude and phase angle.

4. Busbar

Busbars are used to organize system connections and electrically group components. They act as ideal nodes where voltages are defined and power injections/withdrawals are calculated.

In this model, Busbar 0 is associated with the Swing Bus, and Busbar 1 is associated with the Load/PV Bus.

5. Transmission Line (Three-Phase)

A distributed parameter three-phase line representing the electrical connection between the two buses.

It models series impedance and shunt admittance, enabling simulation of real power losses, voltage drops, and reactive power flow between the Swing Bus and the PV Bus.

6. Static Load Bus

This is a secondary bus where an additional static load is placed.

It simulates a consumer receiving power through the transmission line.

The load draws constant active and reactive power, affecting voltage levels and power flow on the line.

7. PV Bus (Generator/Controlled Bus)

The PV Bus represents a generator bus that maintains a specified voltage magnitude by adjusting its reactive power output while supplying a fixed active power level.

It is used to analyse how generation supports voltage regulation within the system.

Conclusion

The 2-Bus Loadflow model provides a simplified yet effective framework for understanding fundamental power-system behavior. It demonstrates the interactions between loads, buses, generators, and transmission lines, making it valuable for educational, analytical, and pre-design studies.